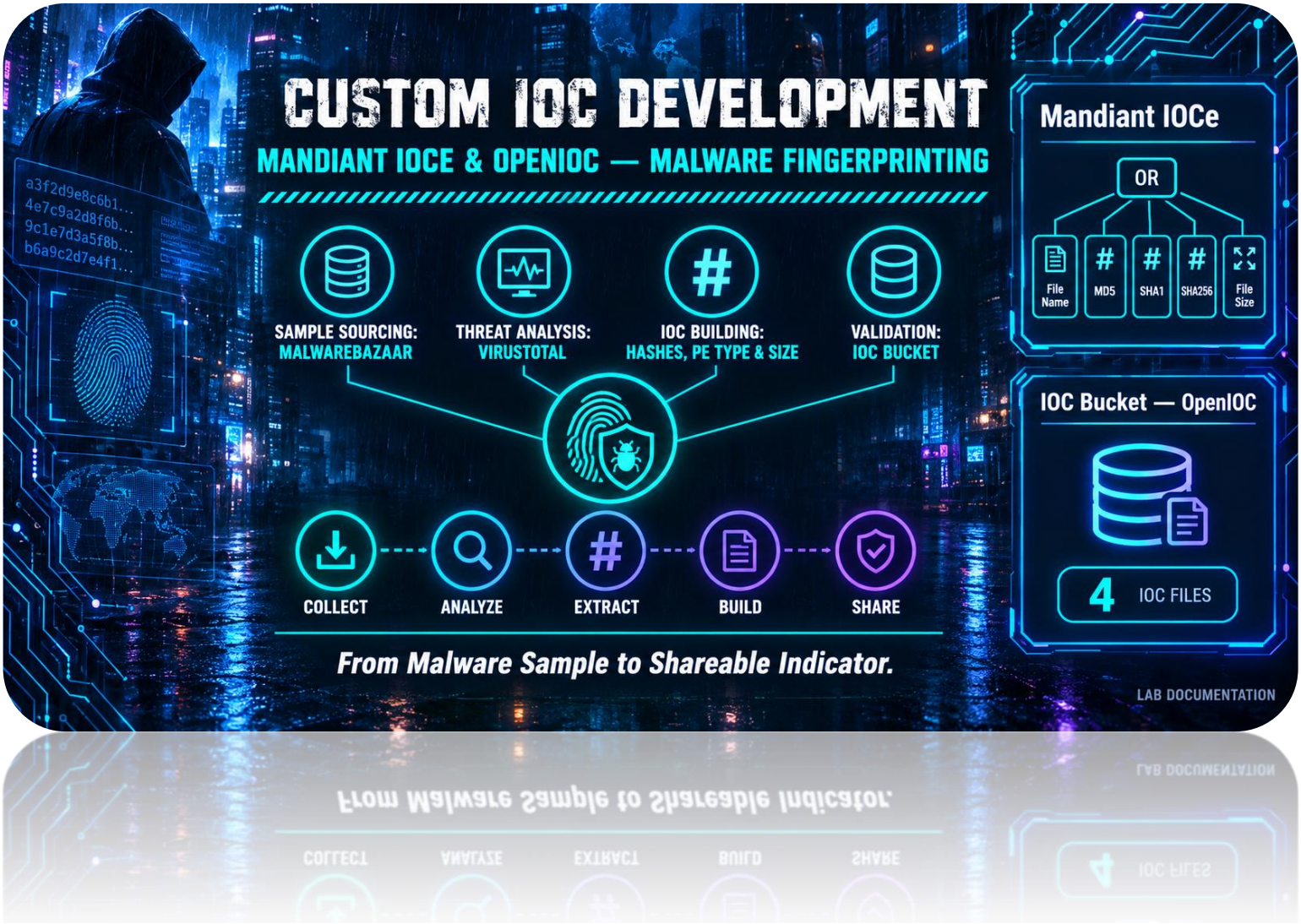


Abdul Wahid

Lab Report

Custom IOC Development – Mandiant IOCe & OpenIOC

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Lab Overview

This report documents the complete process of building custom Indicators of Compromise (IoCs) using Mandiant IOCe 2.2.0 and IOC Bucket (OpenIOC Online Editor). Real malware samples were sourced from MalwareBazaar, analyzed on VirusTotal, and their identifying file properties were encoded into structured .ioc files for use in threat detection and intelligence sharing.

Lab Name	Data Collection through Building Custom IoCs
Author	Analyst Abdul Wahid
Date	April 02, 2026
Primary Tool	Mandiant IOCe 2.2.0
Secondary Tool	IOC Bucket – OpenIOC Online Editor (Beta)
Malware Source	MalwareBazaar (abuse.ch)
Analysis Platform	VirusTotal
IOCs Created	4 custom IOC files (.ioc format)
IOC Directory	C:\Users\micro\Downloads\Lab 7\Lab 7\IOCs

Tool Installation — Mandiant IOCe 2.2.0

Step 2.1 — Setup Wizard Welcome Screen

Mandiant IOCe was installed on the Windows machine. The setup wizard was launched and the Welcome screen was displayed.



Figure 1: Mandiant IOCe Setup Wizard — Welcome Screen

Step 2.2 — Selecting Installation Folder

The installation folder was set to A:\IOC\ and 'Just me' was selected for user scope. Clicked Next to proceed.



Figure 2: Mandiant IOCe — Select Installation Folder (A:\IOC\)

Workflow

The following workflow was followed for each malware sample:

1. Browse MalwareBazaar and select a malware sample
2. Download the sample (ZIP password: infected)
3. Upload hash/file to VirusTotal for analysis and property extraction
4. Open Mandiant IOCe → File → New → Indicator
5. Fill Name, Author, Description fields
6. Right-click OR → Add Item → FileItem → add File Name, MD5, SHA1, SHA256, PE Type, File Size
7. Save the IOC file to the IOCs directory
8. Repeat in IOC Bucket (OpenIOC Online Editor) for validation

Malware 1 — QWnk.exe (Snake Keylogger)

Step 4.1 — VirusTotal Analysis

The QWnk.exe sample was searched on VirusTotal. It was flagged by 24 out of 71 security vendors as malicious. The threat label identified it as trojan.msil/snakekeylogger.

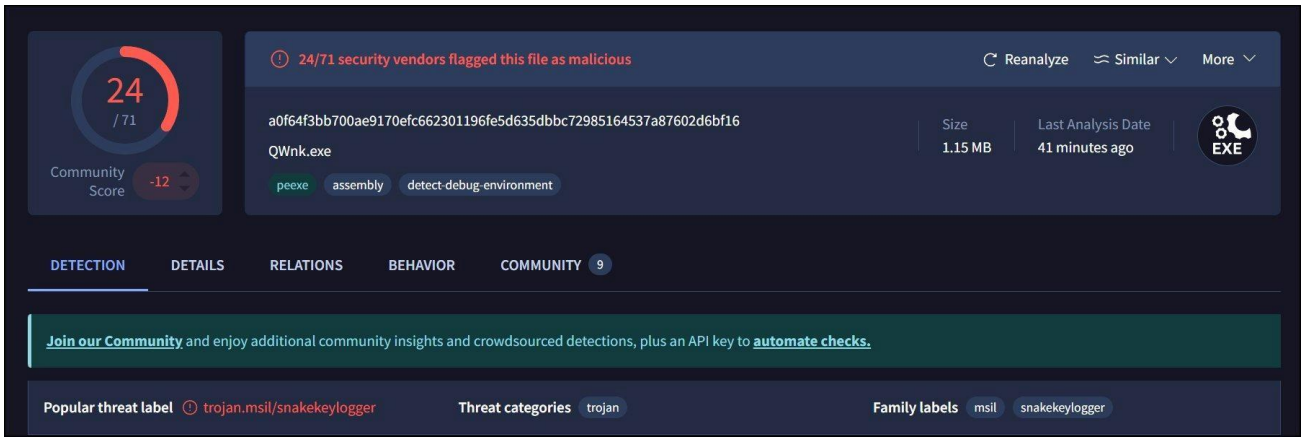


Figure 3: VirusTotal — QWnk.exe flagged by 24/71 vendors as trojan.msil/snakekeylogger

Step 4.2 — Basic Properties (VirusTotal)

The Basic Properties tab on VirusTotal revealed the file hashes, file type, size, and other metadata needed for IOC creation.

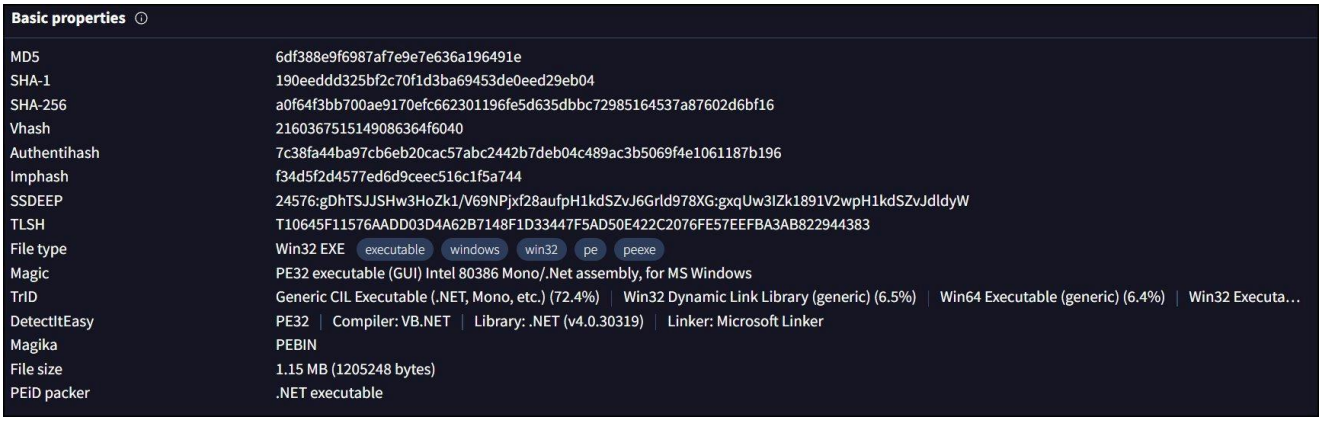


Figure 4: VirusTotal — QWnk.exe Basic Properties (MD5, SHA1, SHA256, File Size, PE Type)

Step 4.3 — Downloading from MalwareBazaar

The QWnk.exe sample was downloaded from MalwareBazaar. The download page shows the SHA256 hash and warns users that the ZIP is password-protected with password 'infected'.

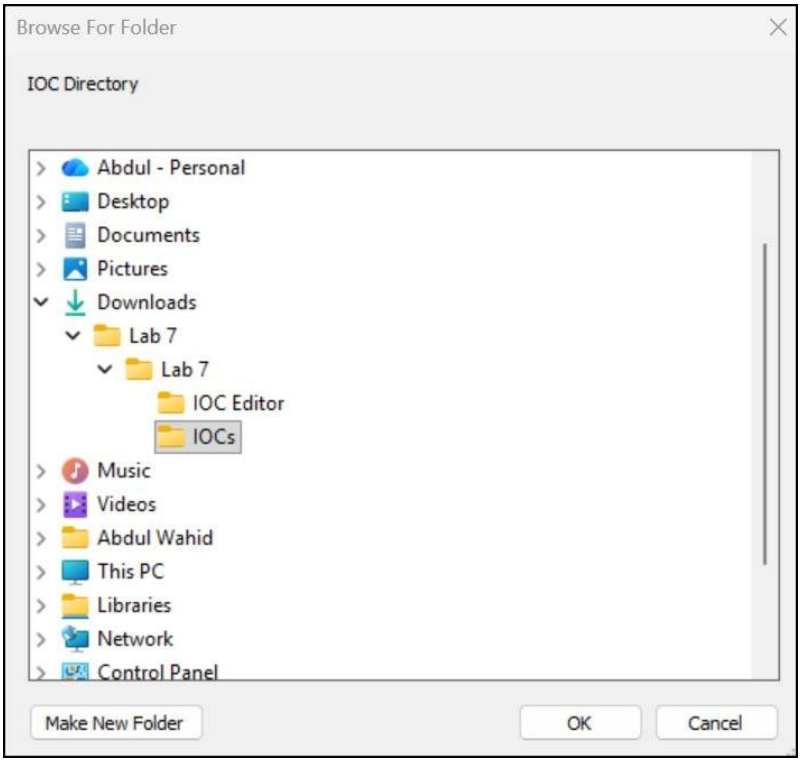


Figure 5: MalwareBazaar — Downloading QWnk.exe sample (SHA256 confirmed)

Step 4.4 — Creating New Indicator in IOCe

In Mandiant IOCe, File → New → Indicator was selected to create a new blank IOC indicator for the QWnk malware.

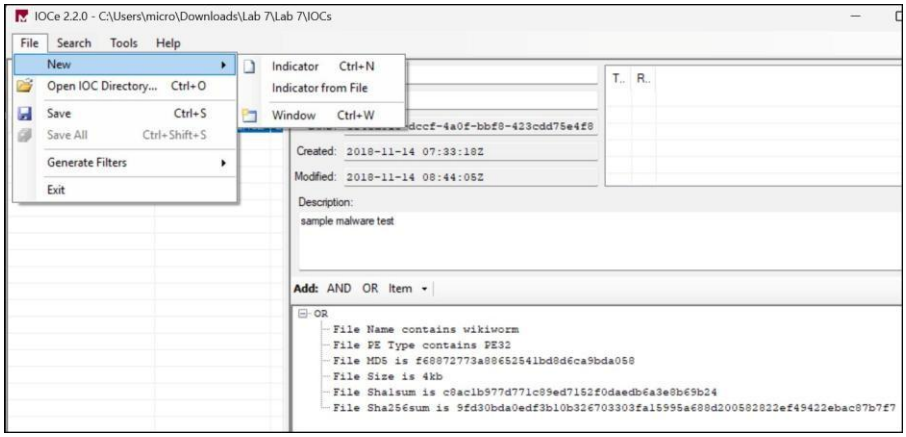


Figure 6: IOCe — File → New → Indicator to create new custom IOC

Step 4.5 — New Unsaved Indicator Created

A new blank indicator named 'New Unsaved Indicator' was created. The GUID was automatically assigned. Basic details (Name, Author, Description) were then filled in.

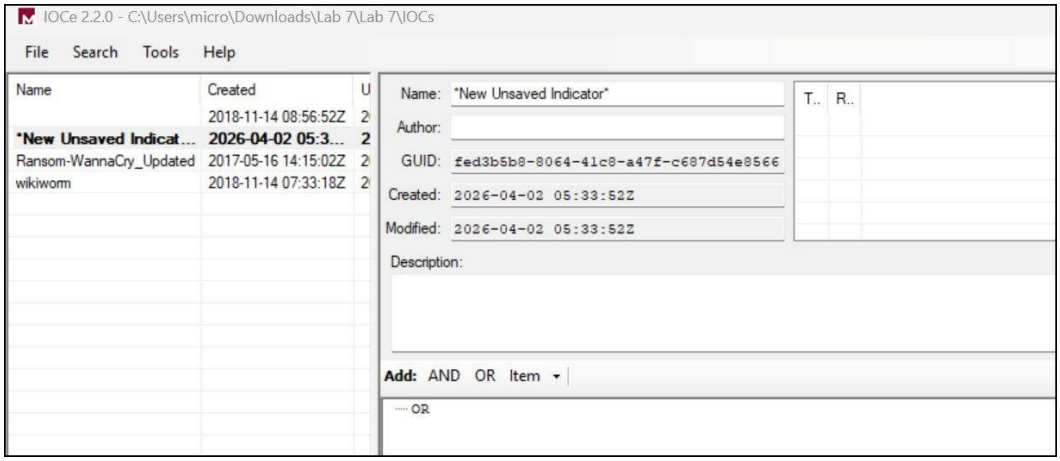


Figure 7: IOCe — New Unsaved Indicator blank form ready for input

Step 4.6 — Adding FileItem Attributes

Right-clicking on '....OR' in the Add section opens the item menu. FileItem was selected, then individual fields like File Name, File MD5, File Sha1sum, File Sha256sum, File PE Type, and File Size were added one by one.

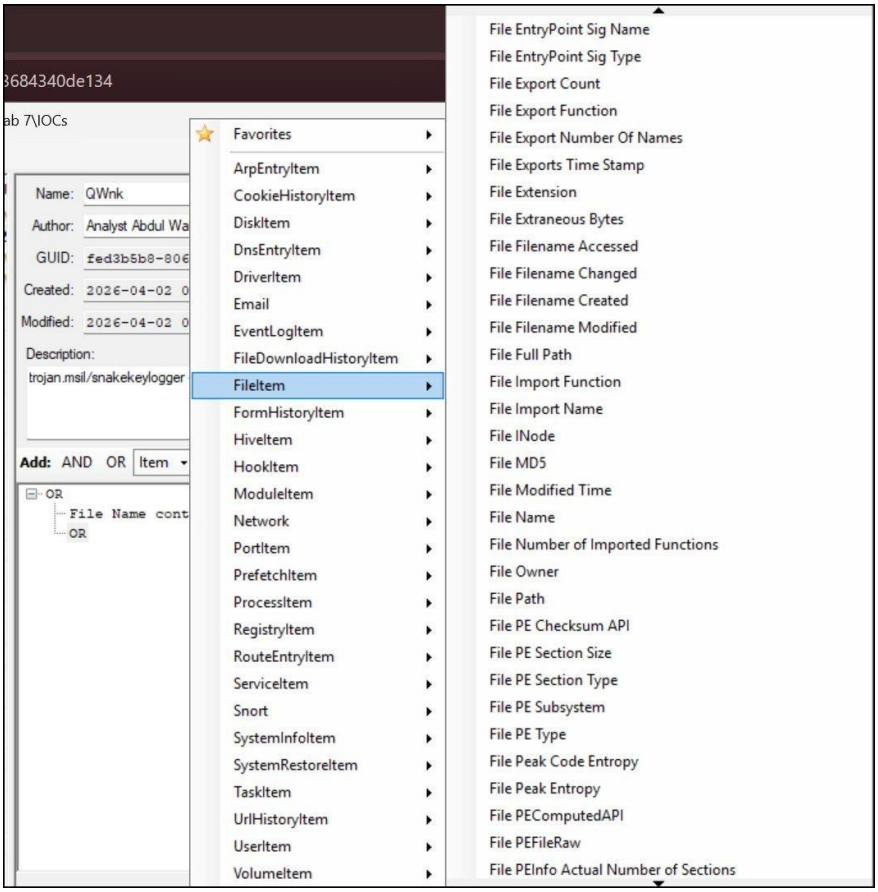


Figure 8: IOCe — Right-click Add Item → FileItem menu showing all available file attributes

Step 4.7 — Completed QWnk IOC

All QWnk.exe attributes were successfully added to the IOC indicator. The OR logic block shows all the file properties that will be used to detect this malware.

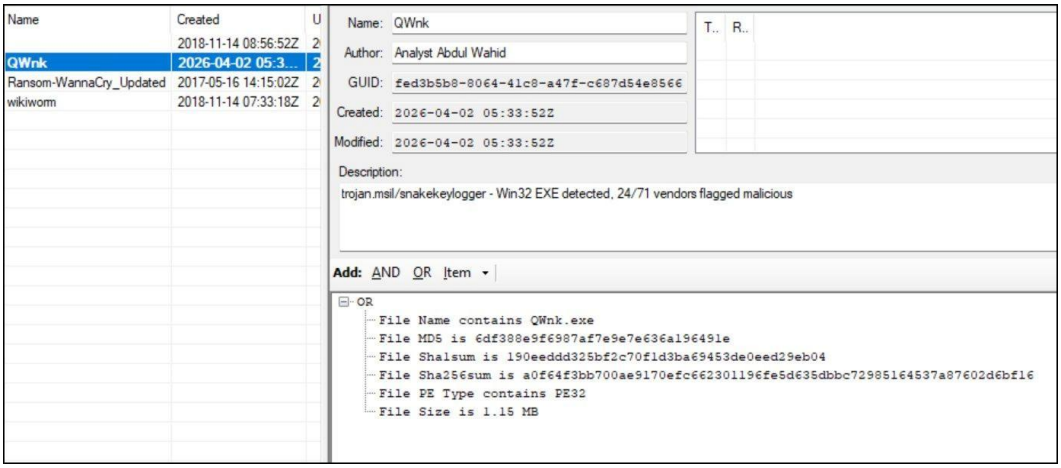


Figure 9: IOCe — QWnk IOC complete with File Name, MD5, SHA1, SHA256, PE Type, File Size

QWnk IOC Details

Field	Value
IOC Name	QWnk
Author	Analyst Abdul Wahid
Threat Label	trojan.msil/snakekeylogger
File Name	QWnk.exe
File Type	Win32 EXE — PE32 executable, .NET assembly
MD5	6df388e9f6987af7e9e7e636a196491e
SHA-1	190eeddd325bf2c70f1d3ba69453de0eed29eb04
SHA-256	a0f64f3bb700ae9170efc662301196fe5d635dbbc72985164537a87602d6bf16
PE Type	PE32
File Size	1.15 MB
Vendors Flagged	24/71

Malware 2 — Mirai DDoS (ELF)

Step 5.1 — VirusTotal Detection

The Mirai DDoS ELF sample was analyzed on VirusTotal. 20 out of 63 vendors flagged it. The threat label identified it as trojan.mirai/ddos targeting Linux systems via CVE-2021-36260 exploit.

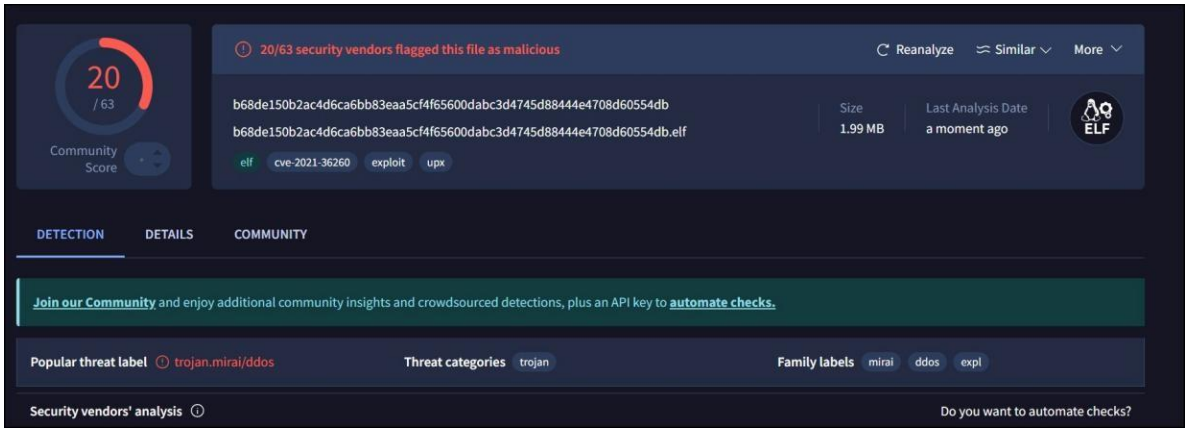


Figure 10: VirusTotal — Mirai DDoS ELF flagged by 20/63 vendors as trojan.mirai/ddos

Step 5.2 — Basic Properties

VirusTotal Basic Properties revealed ELF32 MIPS format, UPX packer, and all required hashes.

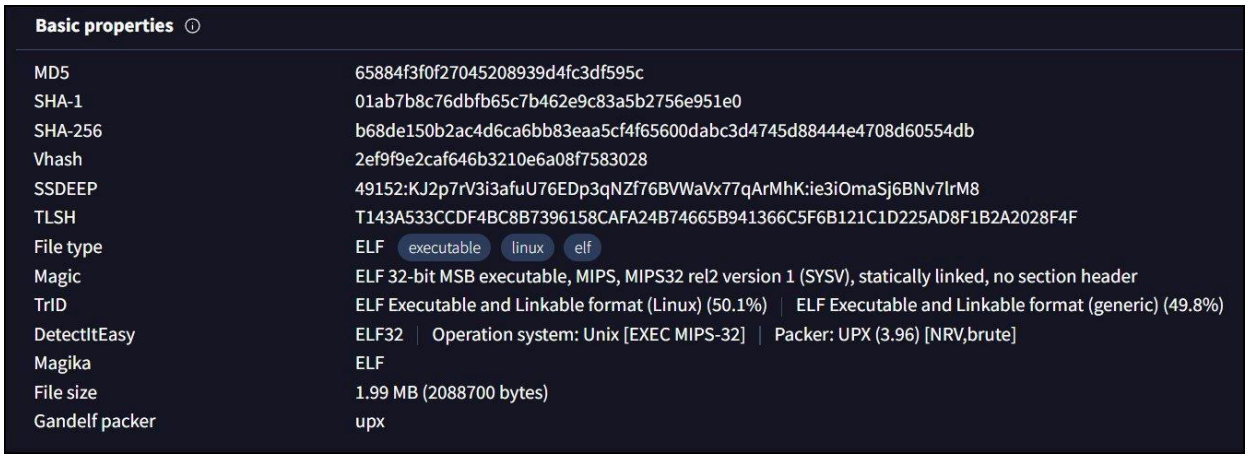


Figure 11: VirusTotal — Mirai DDoS Basic Properties (ELF32, MIPS, UPX packed)

Step 5.3 — Downloading from MalwareBazaar



Figure 12: MalwareBazaar — Downloading Mirai DDoS ELF sample

Step 5.4 — Completed Mirai-DDoS IOC in IOCe

The Mirai-DDoS IOC was created with all file attributes including File Name (the SHA256 hash as filename), MD5, SHA1, SHA256, PE Type (ELF32), and File Size.

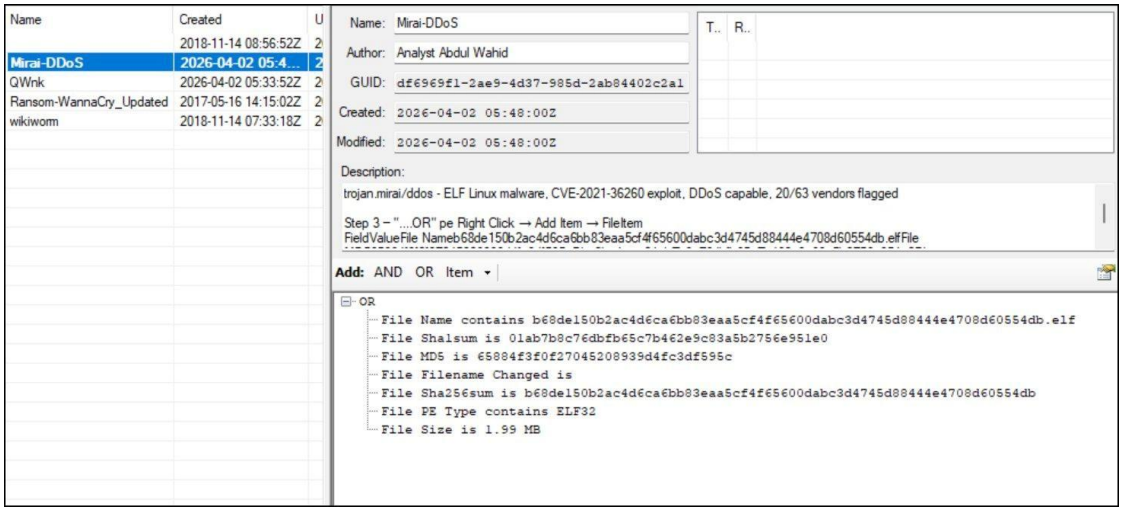


Figure 13: IOCe — Mirai-DDoS IOC complete with all attributes

Mirai-DDoS IOC Details

Field	Value
IOC Name	Mirai-DDoS
Threat Label	trojan.mirai/ddos
Platform	ELF Linux — MIPS32
MD5	65884f3f0f27045208939d4fc3df595c
SHA-1	01ab7b8c76dbfb65c7b462e9c83a5b2756e951e0
SHA-256	b68de150b2ac4d6ca6bb83eaa5cf4f65600dabc3d4745d88444e4708d60554db
PE Type	ELF32
File Size	1.99 MB
CVE	CVE-2021-36260
Vendors Flagged	20/63

Malware 3 — Installer Malware

Step 6.1 — VirusTotal Detection

The Installer.exe sample was flagged by 10 out of 71 vendors. It is a suspicious PE32 installer that executes dropped files and calls WMI.

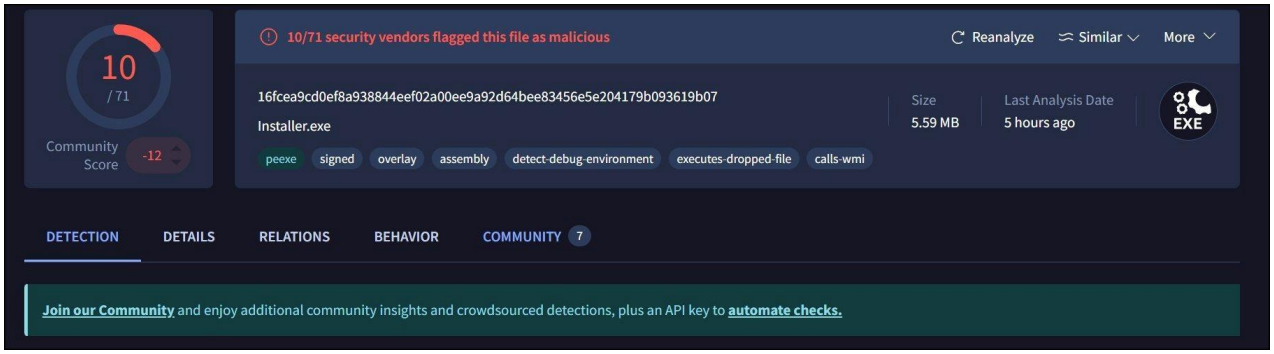


Figure 14: VirusTotal — Installer.exe flagged by 10/71 vendors

Step 6.2 — Basic Properties

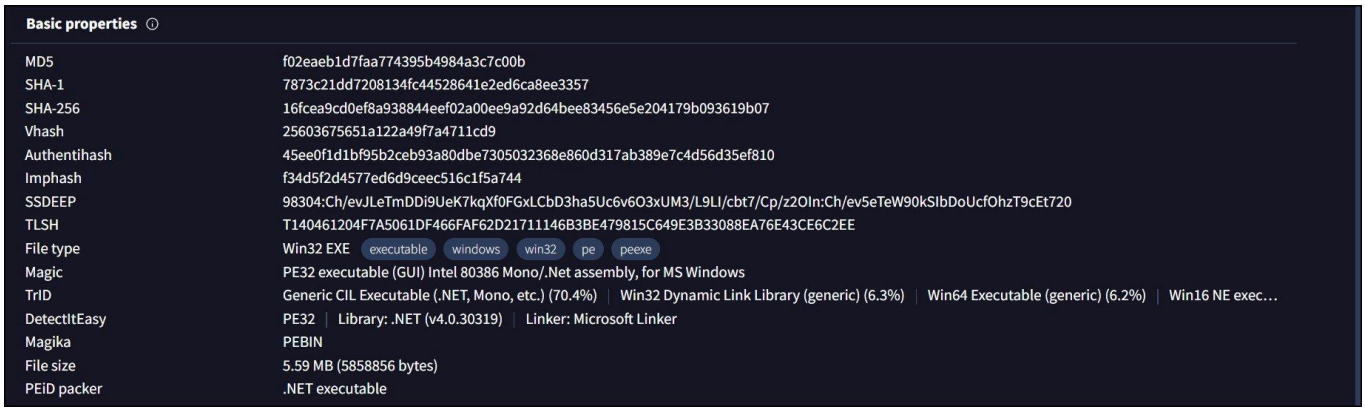


Figure 15: VirusTotal — Installer.exe Basic Properties (PE32, 5.59 MB, .NET)

Step 6.3 — Downloading from MalwareBazaar

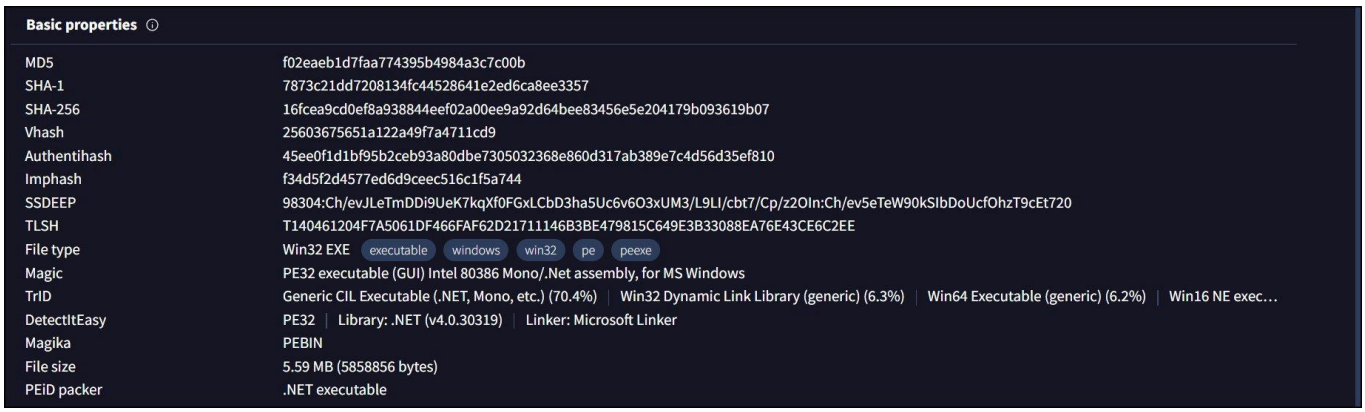


Figure 16: MalwareBazaar — Downloading Installer Malware sample

Step 6.4 — Completed Installer-Malware IOC in IOCe

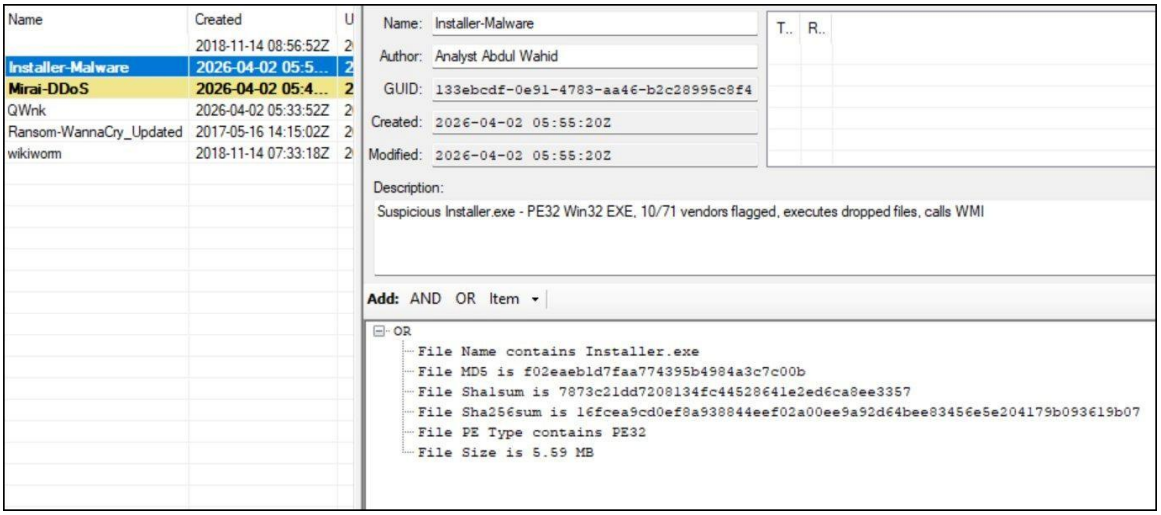


Figure 17: IOCe — Installer-Malware IOC complete with all attributes

Installer Malware IOC Details

Field	Value
IOC Name	Installer-Malware
Threat Label	Suspicious PE32 Installer / Dropper
File Name	Installer.exe
MD5	f02eaeb1d7faa774395b4984a3c7c00b
SHA-1	7873c21dd7208134fc44528641e2ed6ca8ee3357
SHA-256	16fcea9cd0ef8a938844eef02a00ee9a92d64bee83456e5e204179b093619b07
PE Type	PE32
File Size	5.59 MB
Vendors Flagged	10/71

Malware 4 — libsystemd Malware (ELF64)

Step 7.1 — VirusTotal Detection

The .libsystemd.so sample is a Linux ELF64 malware disguised as a system library file. 18 out of 58 vendors flagged it. It uses CVE-2021-41773 exploit and achieves persistence.

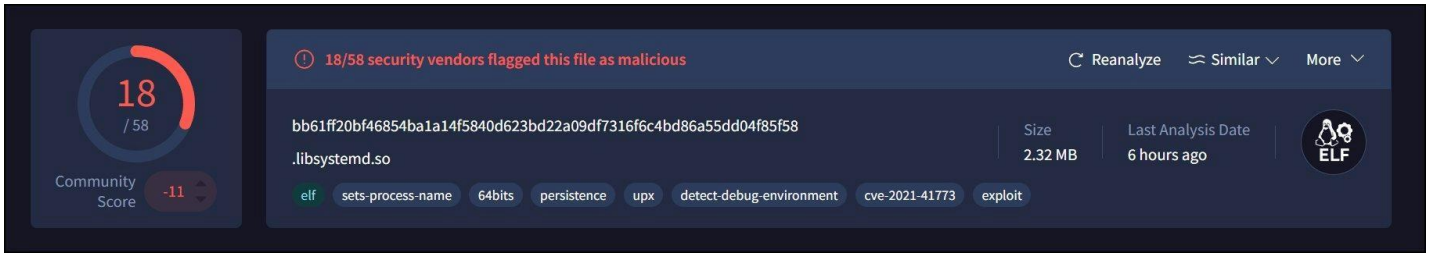


Figure 18: VirusTotal — .libsystemd.so flagged by 18/58 vendors

Step 7.2 — Basic Properties

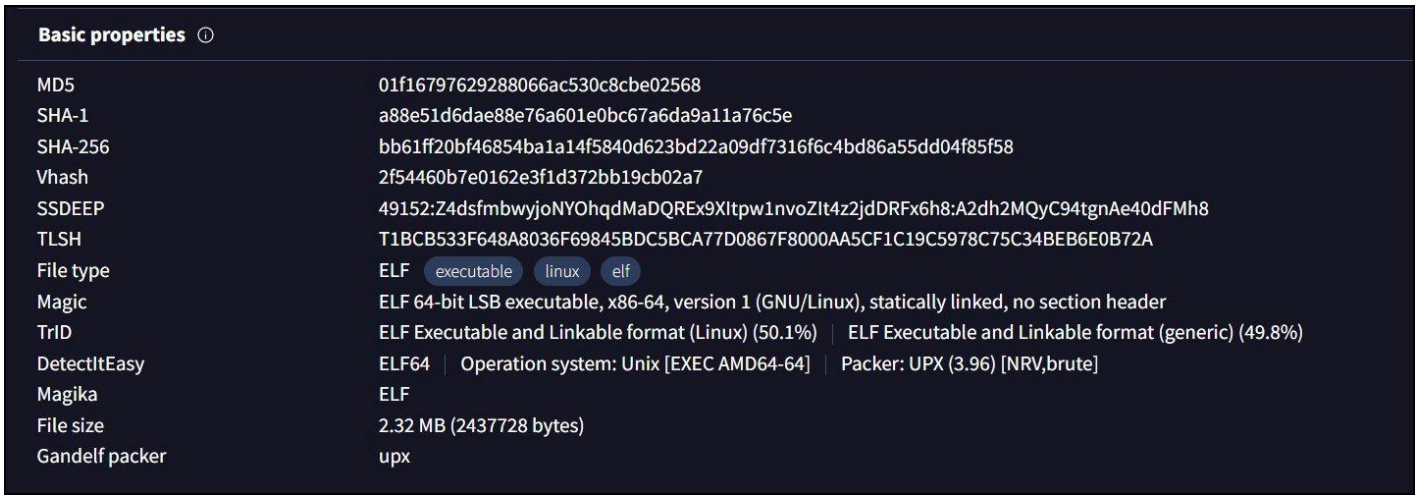


Figure 19: VirusTotal — .libsystemd.so Basic Properties (ELF64, x86-64, UPX packed)

Step 7.3 — Downloading from MalwareBazaar



Figure 20: MalwareBazaar — Downloading .libsystemd.so sample

Step 7.4 — Completed libsystemd-Malware IOC in IOCe

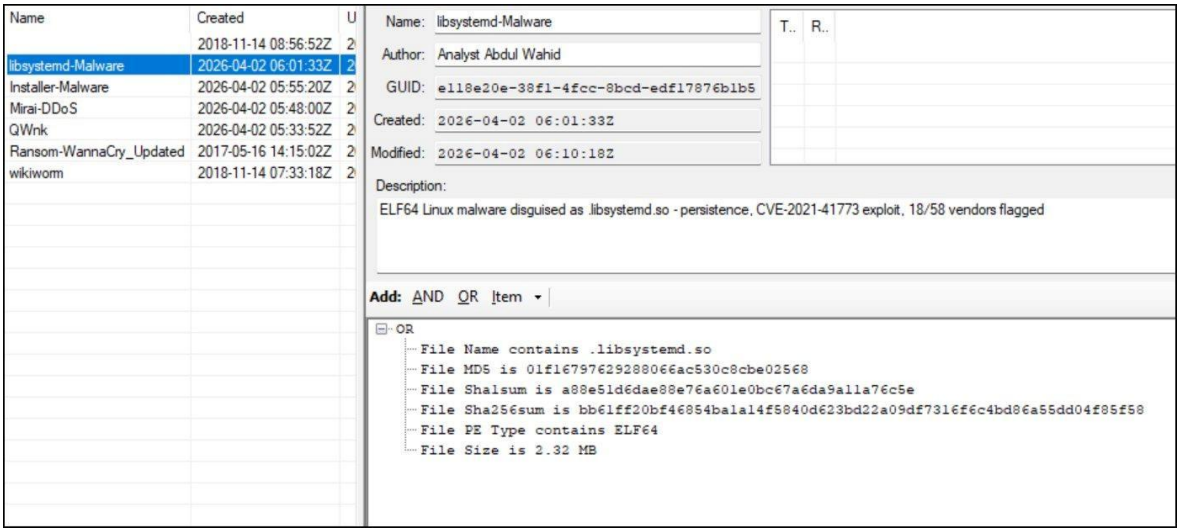


Figure 21: IOCe — libsystemd-Malware IOC complete with all attributes

libsystemd Malware IOC Details

Field	Value
IOC Name	libsystemd-Malware
Threat Label	ELF64 Linux Persistence Malware
File Name	.libsystemd.so
MD5	01f16797629288066ac530c8cbe02568
SHA-1	a88e51d6dae88e76a601e0bc67a6da9a11a76c5e
SHA-256	bb61ff20bf46854ba1a14f5840d623bd22a09df7316f6c4bd86a55dd04f85f58
PE Type	ELF64
File Size	2.32 MB
CVE	CVE-2021-41773
Vendors Flagged	18/58

IOC Bucket — OpenIOC Online Editor Validation

All four IOCs were also created in the IOC Bucket OpenIOC Online Editor (Beta) at https://www.iocbucket.com/openioceditor for validation and online storage. The same attributes were entered using the File Items button.

— installer malware (IOC Bucket)

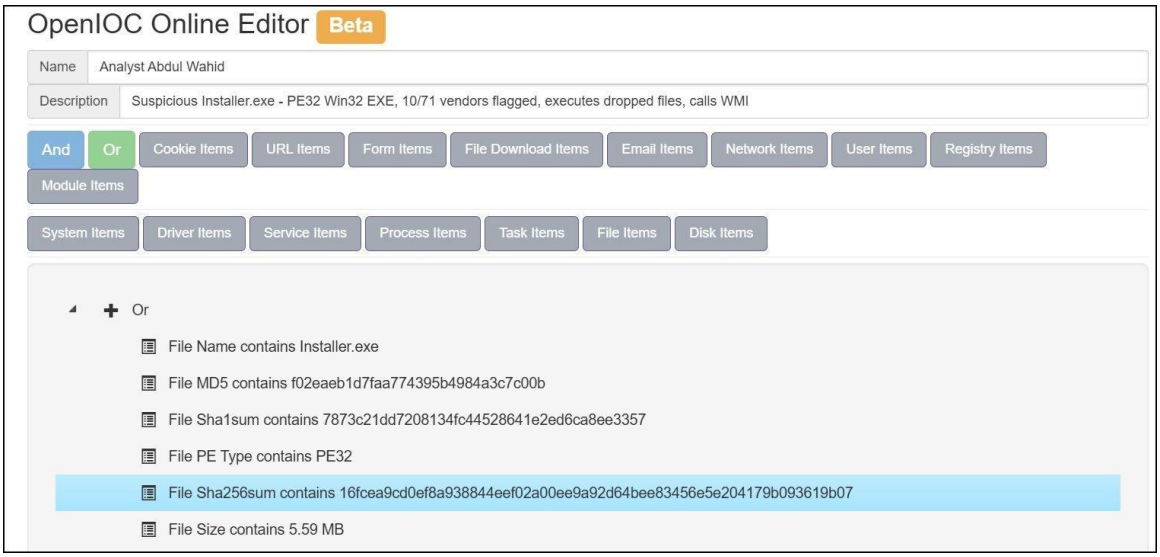


Figure 22: IOC Bucket — installer malware IOC with all file attributes

— Mirai-DDoS (IOC Bucket)

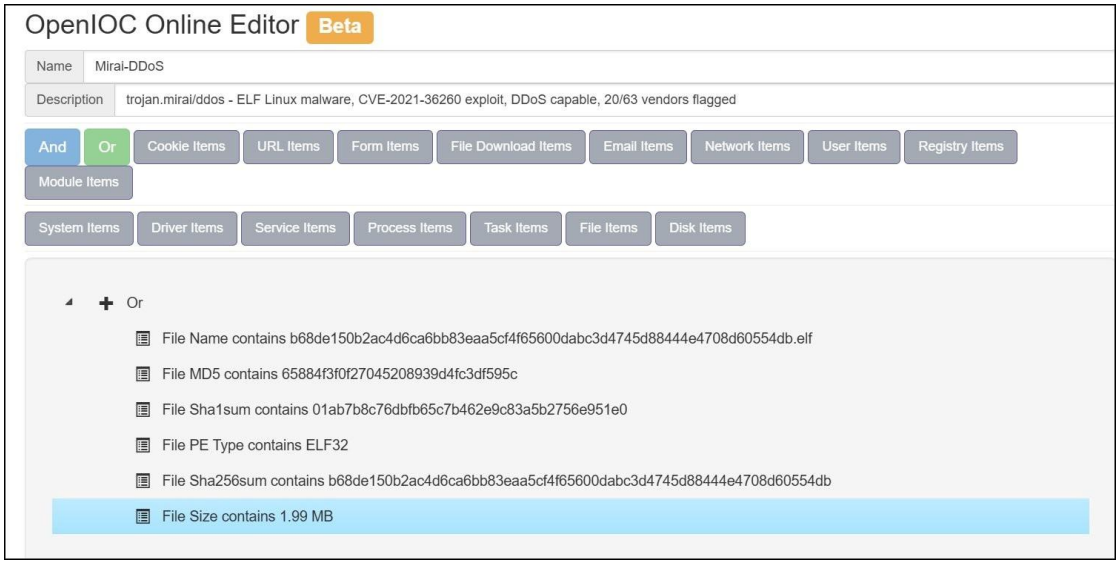


Figure 23: IOC Bucket — Mirai-DDoS IOC with all file attributes

— libsystemd-Malware (IOC Bucket)

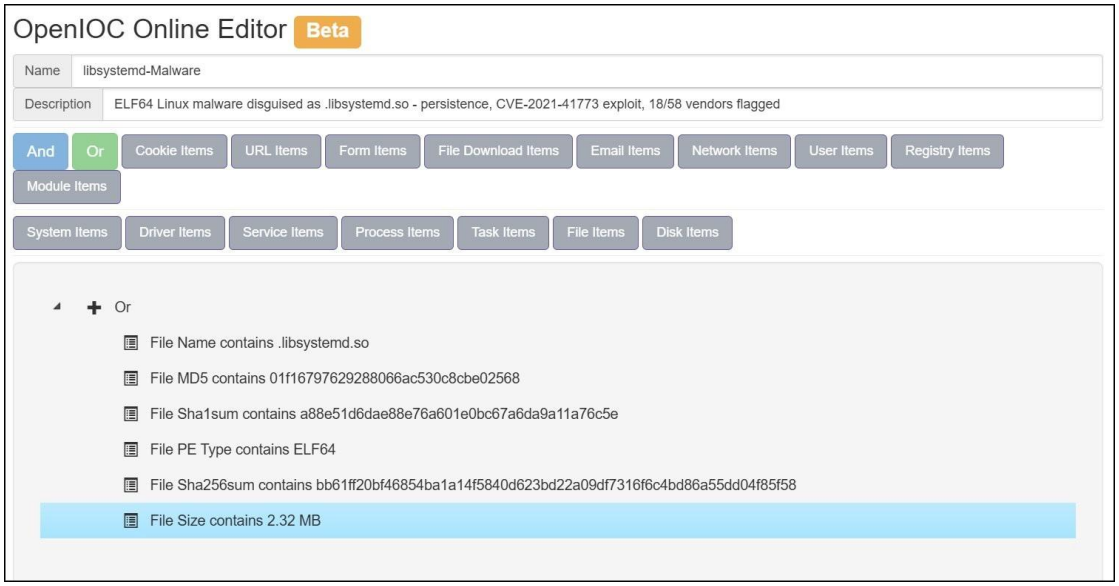


Figure 24: IOC Bucket — libsystemd-Malware IOC with all file attributes

— QWnk (IOC Bucket)

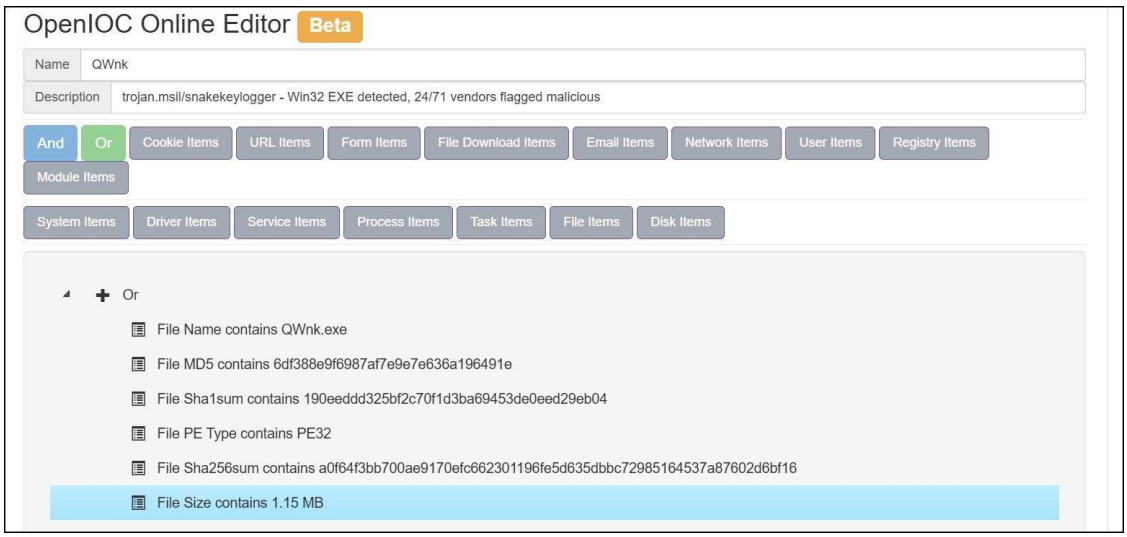


Figure 25: IOC Bucket — QWnk IOC with all file attributes

Saved IOC Files

All four custom IOC files were saved to the IOCs directory in .ioc format. These files can be imported into threat detection tools and shared with threat intelligence communities.

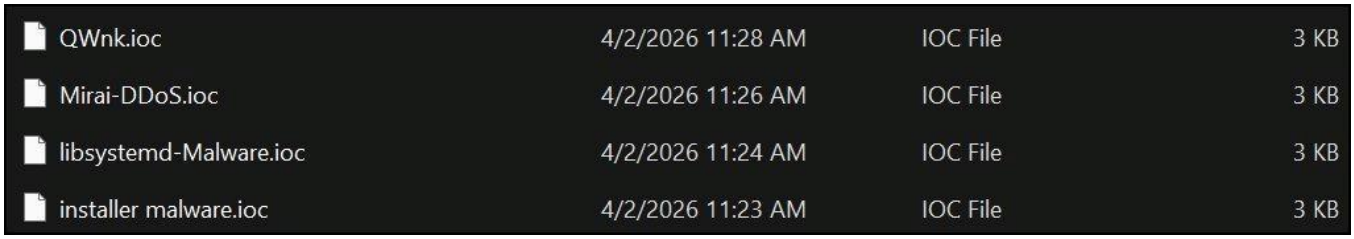


Figure 26: IOCs directory showing all 4 saved .ioc files (QWnk.ioc, Mirai-DDoS.ioc, libsystemd-Malware.ioc, installer malware.ioc)

File Name	Date Created	Type	Size
QWnk.ioc	4/2/2026 11:28 AM	IOC File	3 KB
Mirai-DDoS.ioc	4/2/2026 11:26 AM	IOC File	3 KB
libsystemd-Malware.ioc	4/2/2026 11:24 AM	IOC File	3 KB
installer malware.ioc	4/2/2026 11:23 AM	IOC File	3 KB

Complete IOC Summary

IOC Name	Threat Label	Platform	File Size	Vendors	CVE
QWnk	trojan.msil/snakekeylogger	Win32 PE32	1.15 MB	24/71	—
Mirai-DDoS	trojan.mirai/ddos	ELF32 Linux	1.99 MB	20/63	CVE-2021-36260
Installer-Malware	Suspicious Dropper	Win32 PE32	5.59 MB	10/71	—
libsystemd-Malware	ELF64 Persistence	ELF64 Linux	2.32 MB	18/58	CVE-2021-41773

Conclusion

This lab successfully demonstrated the complete process of building custom Indicators of Compromise (IoCs) using both Mandiant IOCe 2.2.0 and IOC Bucket OpenIOC Online Editor. Four real malware samples were sourced from MalwareBazaar, analyzed using VirusTotal for hash and metadata extraction, and their identifying properties were encoded into structured .ioc files using the OpenIOC framework.

The created IoCs include file fingerprints (MD5, SHA1, SHA256), file type indicators, and size information that can be used by security teams to scan networks and endpoints for the presence of these specific threats. The IOC files are also ready for sharing with threat intelligence communities such as AlienVault OTX and Blueliv Threat Exchange Network.